A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Mineral resources (the lithosphere) - Set 2

Mark scheme

AQA spec 3.2.3; The physical environment; textbook pp. 130-147

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (210 - 12) / 210 x 100 **1** = 94.3% (accept 94%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **recycling** → reprocessing a material to make something new **Do not credit** • A bare wrong answer with no working. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Conserves mineral reserves / extends their lifetime **1** Reduces the environmental impacts of mining (land take, habitat loss, spoil, dust, noise, turbid water) **1** Reduces mineral processing pollution (eg red mud from bauxite refining) **1** Reduces waste disposal impacts / landfill volume **1** Less CO2 / climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
01.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Identification of materials:** slow, labour-intensive to determine composition where it matters (eg alloys, plastic types) **1 Separation of mixed materials:** alloys cannot easily be separated so recycled aluminium cannot be used where pure metal is needed – mixed wastes reduce quality **1** Reduction in quality / contamination (eg paper mixed with food waste) **1 Transport / collection:** small dispersed quantities cannot use bulk transport so costs and energy per tonne are higher (may exceed savings) **1** Lack of consumer cooperation in separating wastes – waste losses to litter and general waste – labour costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
01.4	1 mark for the point below. **1** Designing products so that the materials can be re-used at the end of their lives - easy separation of components and identification of materials - extending the lifetime of mineral use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 146-147; AQA spec 3.2.3.7.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** Lasky's principle - vast quantities exist at low grade – reserves vs resource – COOG affected by price, technology, energy costs • **Improved exploration:** higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • **Low-energy extraction of low-grade ores:** bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • **Previously inaccessible deposits:** polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • **Extending use:** recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • **Evaluation:** environmental impacts vs benefits – energy costs – economics – which strategies are most sustainable – conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **recycling** → reprocessing a material to make something new	9
	• **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Lasky's principle means that as grade falls, the quantity of metal available rises enormously, so depletion of high-grade deposits does not mean running out - it means the cut-off ore grade must fall, which depends on price, technology and energy costs. Better exploration finds what remains: high-resolution multispectral and infrared satellite sensors, gravimetry, magnetometry, seismic and resistivity surveys and portable field analysers narrow the search so expensive trial drilling is targeted. Mechanisation allows deeper underground mines and larger open-cast pits, and previously inaccessible deposits - polymetallic nodules and deep-sea sediments, and Antarctica if its protection were ever lifted - become possible, with serious environmental and legal problems.	
	The most sustainable route to low-grade ores is low-energy extraction: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement of copper, bacterial adsorption of rare earths and polymer adsorption from seawater all avoid the energy of crushing and smelting vast tonnages. Equally important is extending the use of metal already mined: recycling, Cradle-to-Cradle design so products can be disassembled, longer product lifetimes and substitution of scarce metals - the circular economy approach.	
	Each strategy has costs: deep-sea mining and Antarctica risk unique habitats, deeper mines need more energy, recycling is limited by dispersal and mixed alloys. The strategies that mimic natural low-temperature processes and keep metals in use are the most sustainable, and exploration and new mining should be the last resort rather than the first.	
	Weaker answer - Level 1 (3 marks)	
	When high grade ores run out we can mine lower grade ores instead, but this uses more energy. Better technology like satellites can find new deposits. We could mine the deep sea or Antarctica. Bacteria can be used to extract metals from low grade ore. Recycling metals means we do not need to mine as much. These strategies will help secure supplies of metals in the future.	
	What makes the difference	
	• Level 1 names most strategy types but Level 3 explains each (what the surveys measure, why low-temperature extraction matters, what Lasky's principle implies).	
	• Level 3 weighs environmental and energy costs and ranks the strategies; Level 1 offers no evaluation.	
	• Framing the answer with the concept of cut-off ore grade shows understanding rather than recall.	
02.2	1 mark for the point below. **1** All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1 Gravimetry:** variations in gravity caused by differences in the density and mass of the rock below – ore-bearing rocks are usually denser than the surrounding sedimentary deposits **1 Magnetometry:** rocks that are more magnetic than their surroundings, such as the iron ore magnetite and the ores of tungsten and cobalt **1 Seismic surveys:** echoes of sound waves from controlled explosions or a surface vibrator, giving the depth, density and shape of the rock strata **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trial drilling is the most expensive technique per sampling site, so the remote surveys are used first to narrow the search to the most promising ground **1** Drilling is the only technique that brings up actual samples of the rock, so it is used to confirm the composition and purity found by chemical analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reserves are the deposits that can be extracted economically with current technology – the resource base is everything that exists, including deposits that are too low-grade, too deep or too remote to pay **1** A higher price lowers the cut-off ore grade, so deposits that previously could not be mined at a profit become worth extracting and are reclassified as reserves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** All the rock above the ore (the overburden) must be stripped off, so habitat is destroyed over a much larger area than with deep mining **1** Spoil heaps of waste rock occupy land, may be unstable and can be colonised only slowly because they hold few nutrients and little water **1** Dust from blasting, crushing and haulage raises atmospheric particulate levels and settles on surrounding vegetation **1** Noise and vibration from blasting and heavy machinery disturbs wildlife and nearby residents **1** Drainage from the workings and spoil is often acidic and carries dissolved metals, polluting watercourses and groundwater **1** Lowering the water table to keep the pit dry dries out nearby wetlands and wells **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
03.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Topsoil and subsoil stripped at the start are stored separately and replaced in the correct order, so the restored soil has a fertile surface layer **1** The land is re-contoured and spoil heaps are given shallow, stable slopes before being covered with soil and planted **1** The site can be returned to agriculture or forestry, or flooded workings developed as a wetland or water-sport habitat **1** Planning restoration first means the topsoil is saved rather than buried, the final landform is built as mining proceeds, and the cost is included in the economic assessment of the mine **1** A restoration bond or condition in the planning permission makes certain the work is paid for even if the company fails **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 136-142; AQA spec 3.2.3.4.	

Do not write outside the box